

IA Mathematics for Natural Sciences B: Examples Sheet 4 - Derivatives, limits, and continuity

Please send all comments and corrections to jmm232@cam.ac.uk.

A section marked with a (†) contains content that is unique to the Mathematics B course.

Derivatives and curve-sketching

1. (a) From the limit definition, find the derivative of x^n , where n is a non-negative integer.
(b) From the limit definition, show that the derivative of a^x is proportional to a^x . [We define e to be the value of a for which the proportionality constant is one.]
(c) State the *product rule* for derivatives, and derive it directly from the limit definition.
2. State the chain rule, the quotient rule, and the reciprocal rule for derivatives. [These can similarly be derived directly from the limit definition.] Using appropriate rules in each case, find the derivatives of:

$$(a) \log(x), \quad (b) 3^{x^2}, \quad (c) \frac{e^x}{x^3 - 1}, \quad (d) x^3 \log(x^2 - 7), \quad (e) \sqrt{x^3 - e^x \log(x)}.$$

3. By writing each of the following functions in terms of exponentials, compute their derivatives: (a) $\cos(x)$; (b) $\sin(x)$; (c) $\cosh(x)$; (d) $\sinh(x)$; (e) $\tan(x)$; (f) $\tanh(x)$. Learn these derivatives off by heart.
4. Using the reciprocal rule, find the derivatives of: (a) $\cos^{-1}(x)$; (b) $\sin^{-1}(x)$; (c) $\cosh^{-1}(x)$; (d) $\sinh^{-1}(x)$; (e) $\tan^{-1}(x)$; (f) $\tanh^{-1}(x)$. Learn these derivatives off by heart. Now, by first expressing $\sin^{-1}(x)$ and $\sinh^{-1}(x)$ as logarithms instead, then differentiating, verify your results for (b) and (d).
5. Let $y = y(x)$ be a function of x . Using the reciprocal rule and the chain rule, show that:

$$\frac{d^3 x}{dy^3} = - \left(\frac{dy}{dx} \right)^{-4} \frac{d^3 y}{dx^3} + 3 \left(\frac{dy}{dx} \right)^{-5} \left(\frac{d^2 y}{dx^2} \right)^2,$$

and verify this result when $y = e^{2x}$.

6. Explain what is meant by *implicit differentiation*, and why it is called *implicit*. Using: (a) implicit differentiation; (b) the reciprocal rule, find dy/dx given $y + e^y \sin(y) = 1/x$.
7. State what it means for a function to be *even* or *odd*, and the graphical significance of these definitions. Decide whether the following functions are even, odd, both, or neither:

$$(a) x, \quad (b) \sin(x), \quad (c) e^x, \quad (d) \sin\left(\frac{\pi}{2} - x\right), \quad (e) |x| \cos(x), \quad (f) \sqrt{x}, \quad (g) 2, \quad (h) 0, \quad (i) \log \left| \frac{1+x}{1-x} \right|.$$

8. Give a list of things one should consider when sketching the graph of a single-variable function, $y = f(x)$. Sketch the graphs of the following functions, justifying your sketch in each case:

$$(a) (x-3)^3 + 2x, \quad (b) \frac{x}{1+x^2}, \quad (c) \frac{x^2+3}{x-1}, \quad (d) xe^x, \quad (e) \frac{\log(x)}{1+x}, \quad (f) \frac{1}{1-e^x}, \quad (g) e^x \cos(x).$$

(†) The Leibniz derivative rule

9. Using induction, prove the Leibniz derivative rule for the n th derivative of a product. Hence, calculate: (i) the third derivative of $\log^2(x)$; (ii) the 100th derivative of $x^2 e^x$.
10. Show that one solution of the differential equation $Z'' + xZ' + (n+1)Z = 0$ is given by $Z = \frac{d^n}{dx^n}(e^{-x^2/2})$.

(†) Limits and L'Hôpital's rule

11. Give the formal ϵ, δ definition of the phrase ' $f(x) \rightarrow l$ as $x \rightarrow x_0$ '. Draw a diagram to explain your definition. Using the formal ϵ, δ definition, construct proofs that:

$$(a) x^2 \rightarrow 0 \text{ as } x \rightarrow 0, \quad (b) 4x^3 \rightarrow 0 \text{ as } x \rightarrow 0, \quad (c) x^2 \sin(1/x) \rightarrow 0 \text{ as } x \rightarrow 0.$$

12. In practice, we do not use the formal ϵ, δ definition when evaluating limits - instead, mathematicians prove basic '*laws of limits*' from the definition, and use them instead. State these laws (regarding sums, products, quotients, powers, and compositions of functions) making it clear under what conditions the laws apply (take care to identify the 'indeterminate forms'). Use these laws to evaluate:

$$(a) \lim_{x \rightarrow 0} \frac{x+1}{2-x^2}, \quad (b) \lim_{x \rightarrow \infty} \sin\left(\frac{x^2+x+1}{3x^2-4}\right), \quad (c) \lim_{x \rightarrow 0} \left(\exp\left(\frac{x^4-1}{x^4+1}\right)\right)^{1/x^2}, \quad (d) \lim_{x \rightarrow \infty} (\sqrt{x^2+7x}-x).$$

13. State *L'Hôpital's rule* for evaluating limits of differentiable functions, carefully specifying the conditions under which it is valid. Give also a geometric explanation for the validity of the rule. Assuming $\alpha > 0$ throughout, use L'Hôpital's rule (where appropriate) to evaluate the limits of the following functions both: (i) as $x \rightarrow 0^+$ (a *one-sided limit*), and (ii) as $x \rightarrow \infty$:

$$(a) x^\alpha \log(x), \quad (b) x^{-\alpha} \log(x), \quad (c) x^\alpha e^{-x}, \quad (d) x^{-\alpha} e^x, \quad (e) \sin(\alpha x)/x.$$

14. Using L'Hôpital's rule, evaluate the following 'power law' limits:

$$(a) \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^x, \quad (b) \lim_{x \rightarrow \infty} \log^{1/x}(x), \quad (c) \lim_{x \rightarrow 0^+} x^x, \quad (d) \lim_{x \rightarrow \infty} x^{1/x}.$$

In part (a), if we use $1/x \rightarrow 0$ as $x \rightarrow \infty$ to deduce $\lim_{x \rightarrow \infty} (1 - 1/x)^x = \lim_{x \rightarrow \infty} (1 + 0)^x = 1$, why do we obtain the wrong answer?

15. Determine the limit $\lim_{x \rightarrow \infty} \left(1 + a^x + \left(\frac{a^2}{2}\right)^x\right)^{1/x}$, where $a \geq 0$.

(†) Continuity of functions

16. (a) Give the formal ϵ, δ definition of the phrase ' $f(x)$ is continuous at the point $x = x_0$ '. Draw a diagram to explain your definition. Explain why the definition is the same as $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.
(b) Show that if a function is differentiable at a point $x = x_0$, then it is continuous at $x = x_0$. Is the converse true?

17. Consider the function:

$$f(x) = \begin{cases} \exp(-1/x^2), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Sketch $f(x)$. Is the function continuous, and/or differentiable, at $x = 0$? Justify your answers.

18. Consider the functions:

$$f(x) = \begin{cases} |x|^p \sin(1/x), & x \neq 0, \\ 0, & x = 0, \end{cases} \quad g(x) = \begin{cases} |x|^q \sin(\pi(\sin(1/x))), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

- (a) For what values of p is $f(x)$ continuous at $x = 0$? For those values, give a proof using the ϵ, δ definition. For what values of p is $f(x)$ differentiable at $x = 0$?
(b) For what values of q is $g(x)$ continuous, and/or differentiable, at $x = 0$?